



# **034IN - FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL**

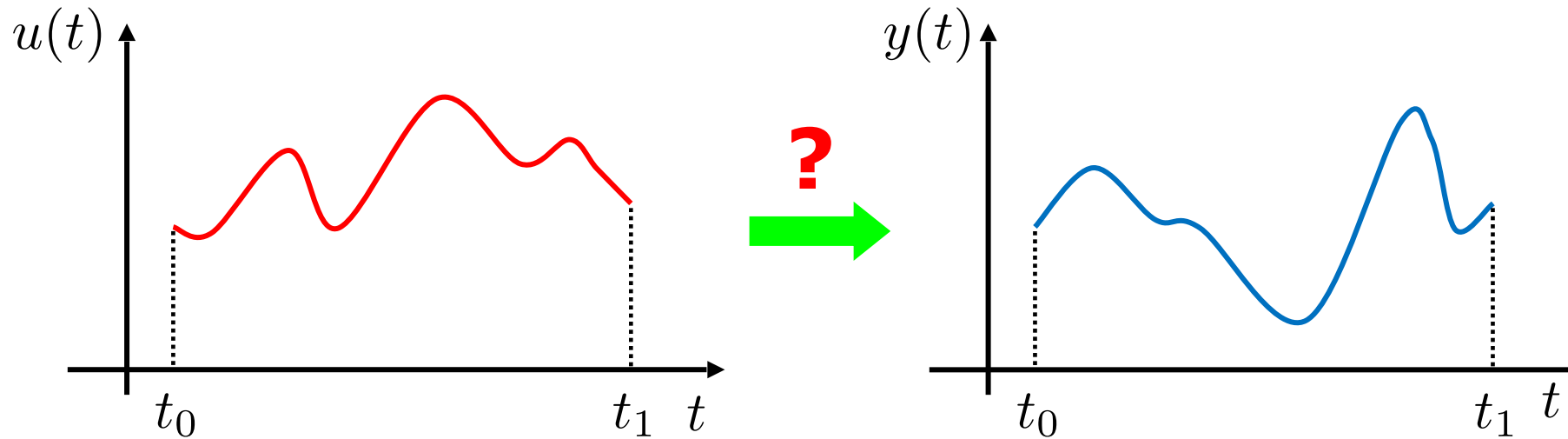
**A.Y. 2024-2025**

**Part II: Fundamentals of Systems Theory**

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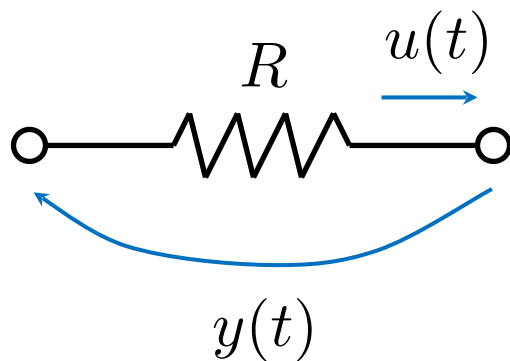
# What is the meaning of “Dynamic”?



Can  $y(t)$  be determined only through the knowledge of  $u(t)$  ?

**If the answer is “no,” then the system is a dynamic one**

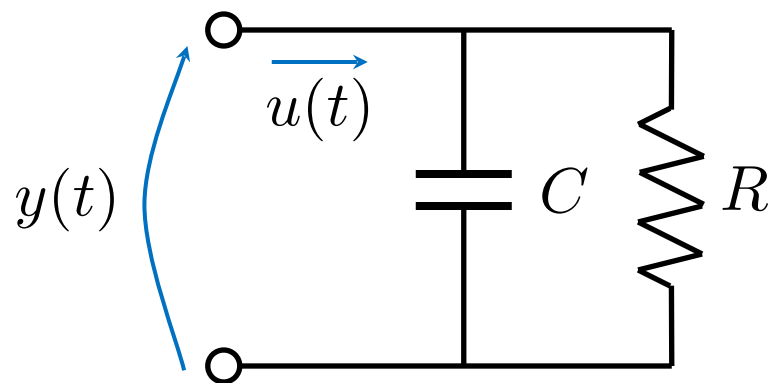
## Example 1



$$y(t) = R \cdot u(t)$$

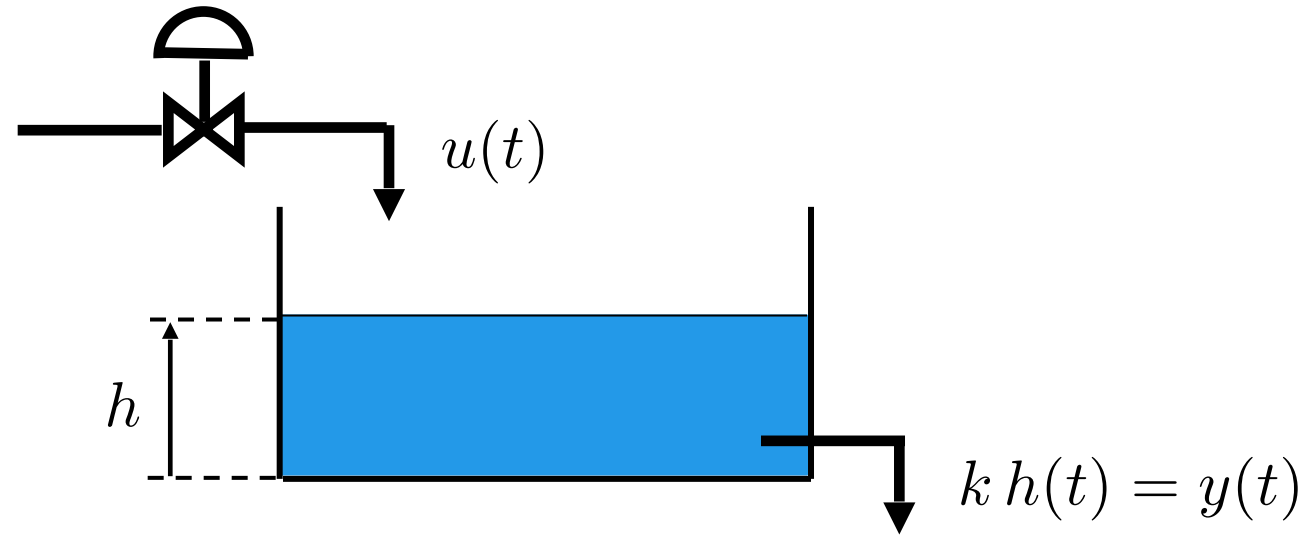
Non dynamic

## Example 2



$$\left. \begin{array}{l} u(t), t \in [t_0, t_1] \\ y(t_0) \end{array} \right\} \rightarrow \text{Dynamic}$$
$$y(t), t \in [t_0, t_1]$$

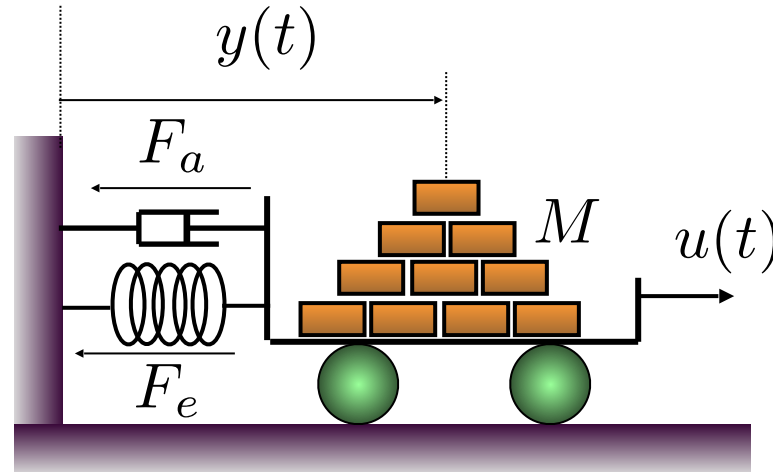
## Example 3



$$\left. \begin{array}{l} u(t), t \in [t_0, t_1] \\ h(t_0) \end{array} \right\} \longrightarrow y(t), t \in [t_0, t_1]$$

Dynamic

## Example 4



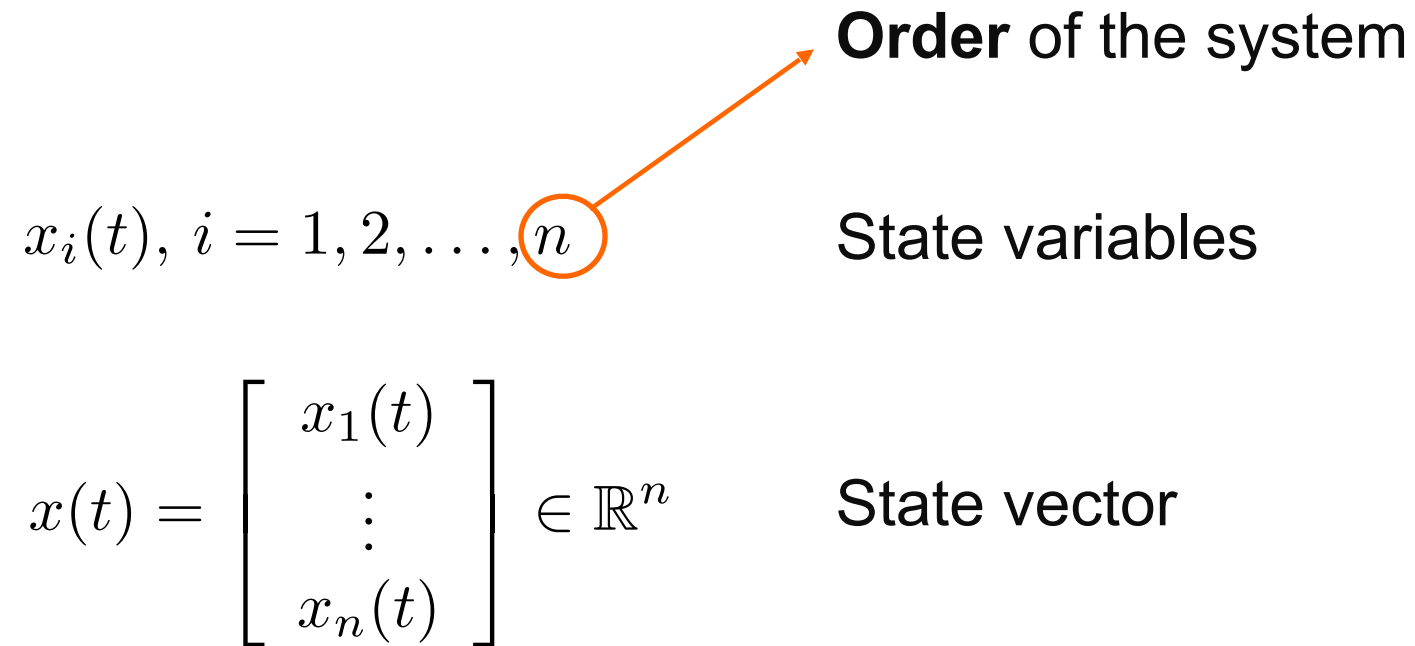
$$\left. \begin{array}{l} u(t), t \in [t_0, t_1] \\ y(t_0) \\ \dot{y}(t_0) \end{array} \right\} \longrightarrow y(t), t \in [t_0, t_1]$$

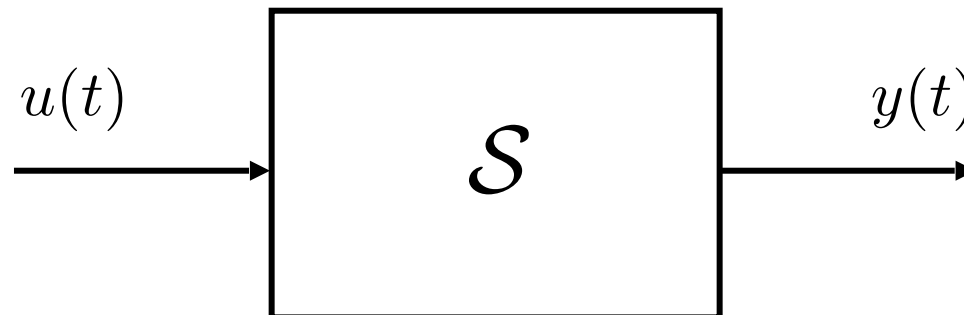
Dynamic

# Continuous-time Dynamic Systems

## Analysis and Properties

Variables that must be known at time  $t_0$  to determine  $y(t), t \geq t_0$  from  $u(t), t \geq t_0$





$$u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_m(t) \end{bmatrix} \in \mathbb{R}^m$$

Input

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix} \in \mathbb{R}^n$$

State

$$y(t) = \begin{bmatrix} y_1(t) \\ \vdots \\ y_p(t) \end{bmatrix} \in \mathbb{R}^p$$

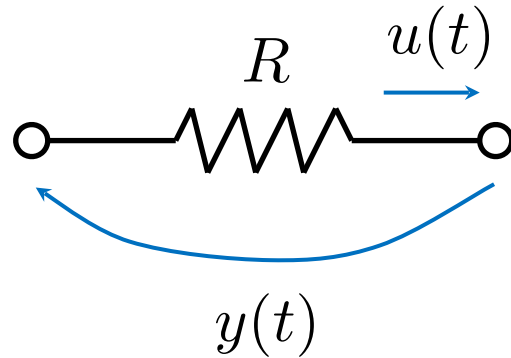
Output

↳  $\begin{cases} \dot{x}(t) = f[x(t), u(t), t] \\ y(t) = g[x(t), u(t), t] \end{cases}$  where  $f[x(t), u(t), t] := \begin{bmatrix} f_1[x(t), u(t), t] \\ \vdots \\ f_n[x(t), u(t), t] \end{bmatrix}$

State equations

- **Strictly proper** dynamic system:
  - ➔ if function  $g(\cdot)$  does not explicitly depend on input
- **Time-invariant** (or **stationary**) dynamic system:
  - ➔ if functions  $f(\cdot), g(\cdot)$  do not explicitly depend on time
- **Linear** dynamic system:
  - ➔ if functions  $f(\cdot), g(\cdot)$  depend linearly on  $x, u$
- **Single Input Single Output (SISO)** dynamic system:
  - ➔ if  $m = p = 1$
- **Multi-Input Multi-Output (MIMO)** dynamic system:
  - ➔ if  $m > 1$  and/or  $p > 1$

# State Equations for Example 1



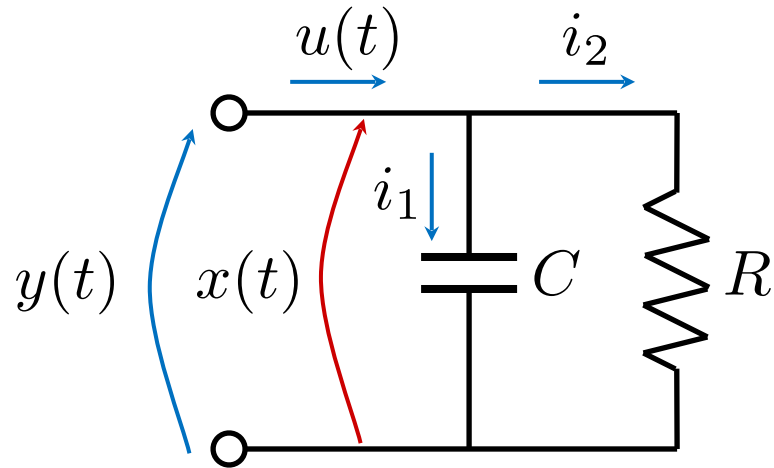
$$y(t) = R \cdot u(t)$$

Non dynamic



**No need** to introduce state variables

# State Equations for Example 2

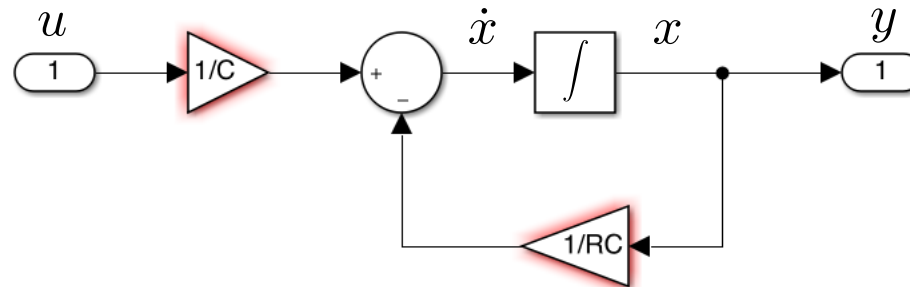


From basic physics and electrical circuit theory:

$$\begin{cases} C\dot{x} = i_1 \\ y = x = Ri_2 \\ u = i_1 + i_2 \end{cases}$$

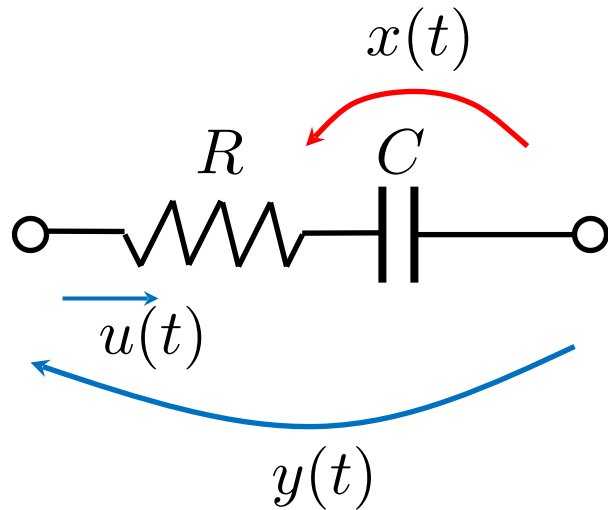
$$x = Ri_2 = R(u - i_1) \quad \longrightarrow \quad i_1 = u - \frac{1}{R}x \quad \longrightarrow \quad \dot{x} = \frac{1}{C}i_1 = \frac{1}{C} \left( u - \frac{1}{R}x \right)$$

$$\begin{cases} \dot{x}(t) = -\frac{1}{RC}x + \frac{1}{C}u = f(x, u) \\ y = x = g(x) \end{cases}$$



- First order
- Linear
- Time-invariant
- Strictly proper
- SISO

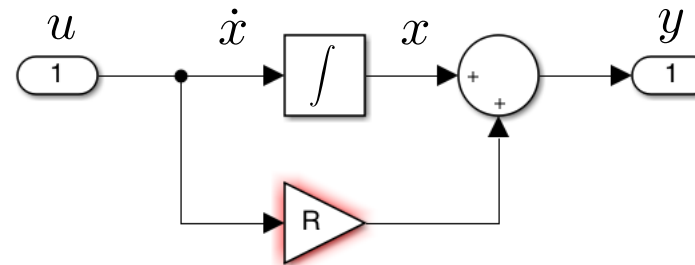
# State Equations for Example 2-bis



From basic physics and electrical circuit theory:

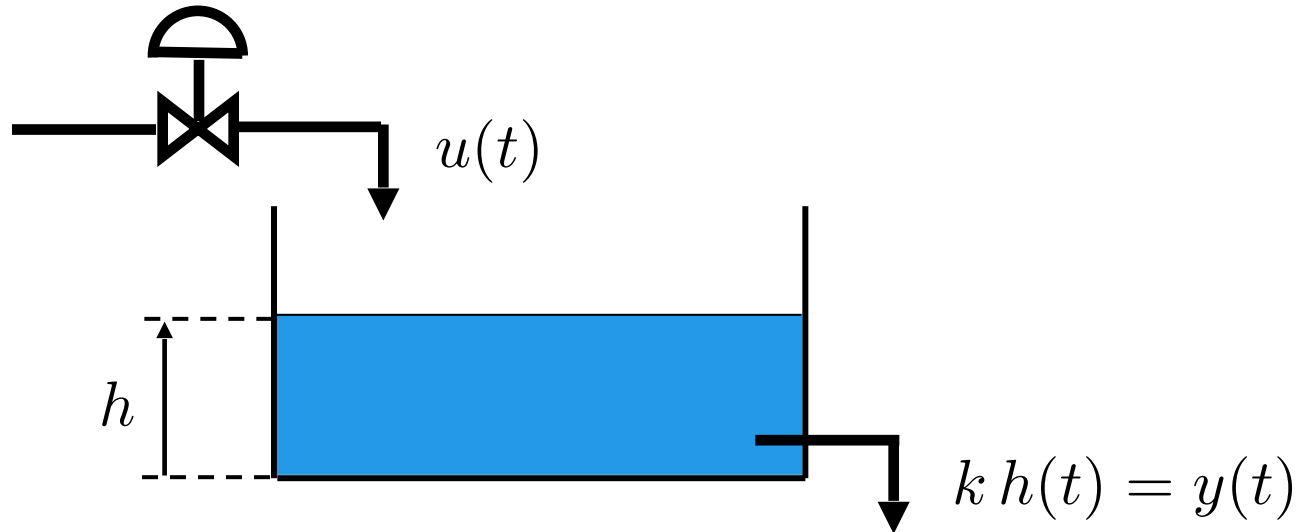
$$\begin{cases} C\dot{x} = u \\ y = x + Ru \end{cases}$$


$$\begin{cases} \dot{x}(t) = \frac{1}{C} u = f(x, u) \\ y = x + Ru = g(x, u) \end{cases}$$



- First order
- Linear
- Time-invariant
- Non strictly proper
- SISO

# State Equations for Example 3

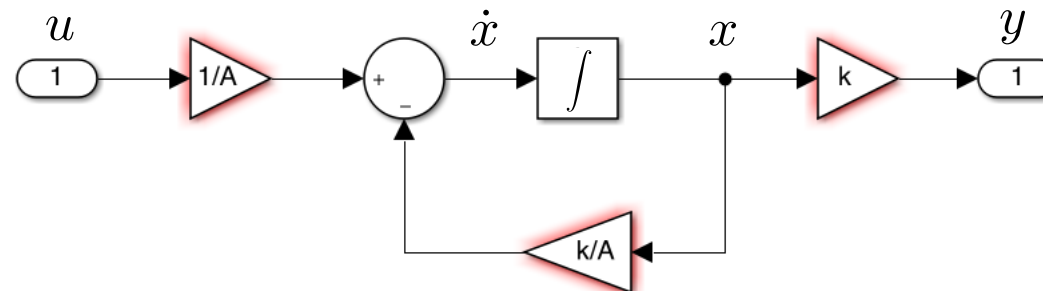


From elementary physics:

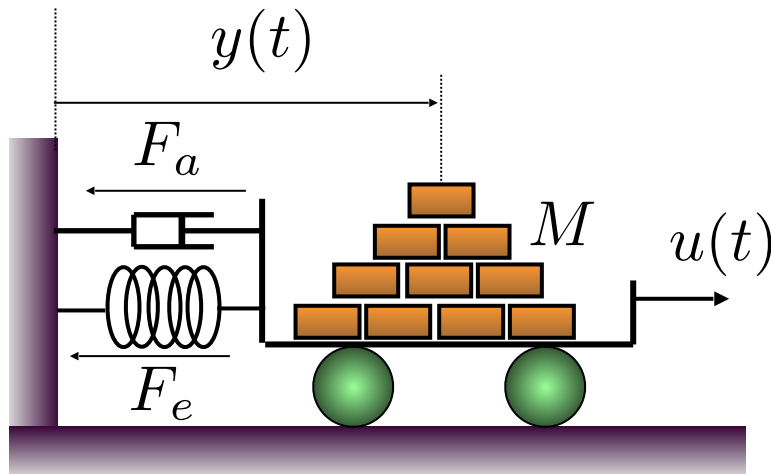
$$\begin{cases} A \dot{x} = u - k x \\ y = k x \end{cases}$$

➡ 
$$\begin{cases} \dot{x}(t) = -\frac{k}{A} x + \frac{1}{A} u = f(x, u) \\ y = k x = g(x) \end{cases}$$

- First order
- Linear
- Time-invariant
- Strictly proper
- SISO



# State Equations for Example 4



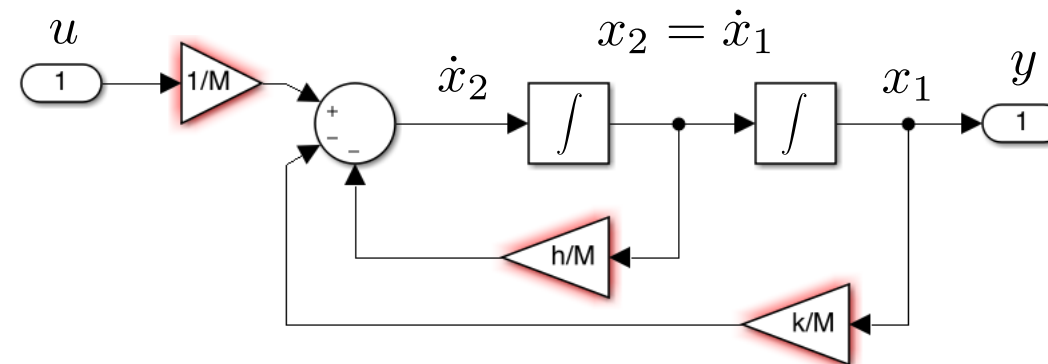
From elementary physics:

$$\begin{cases} F_e = ky \\ F_a = h\dot{y} \\ M\ddot{y} = u - ky - h\dot{y} \end{cases}$$

Letting:

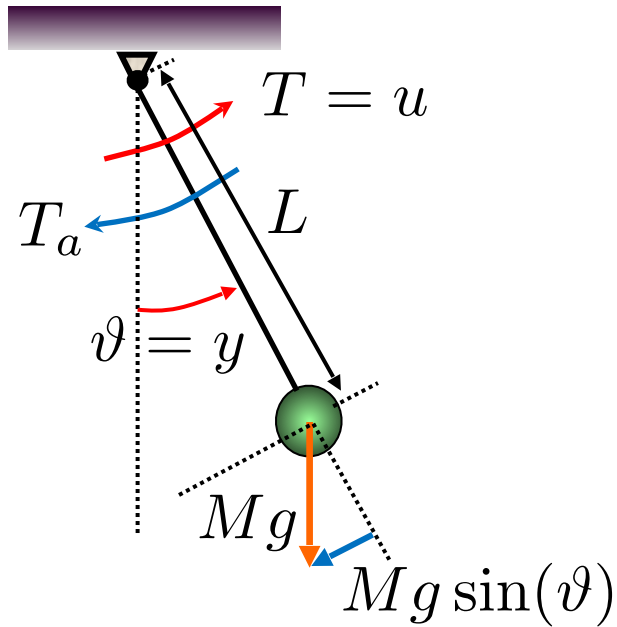
$$\begin{cases} x_1 := y \\ x_2 := \dot{y} \end{cases}, \quad x := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{cases} \dot{x}_1 = \dot{y} = x_2 = f_1(x, u) \\ \dot{x}_2 = \ddot{y} = -\frac{k}{M}x_1 - \frac{h}{M}x_2 + \frac{1}{M}u = f_2(x, u) \\ y = x_1 = g(x) \end{cases}$$



- Second order
- Linear
- Time-invariant
- Strictly proper
- SISO

## Example 5



From elementary physics:

$$\begin{cases} T_a = h\dot{\vartheta} \\ J\ddot{\vartheta} = u - h\dot{\vartheta} - MgL \sin(\vartheta) \\ J = ML^2 \end{cases}$$

Letting:

$$\begin{cases} x_1 := \vartheta \\ x_2 := \dot{\vartheta} \end{cases}, \quad x := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

➡ 
$$\begin{cases} \dot{x}_1 = \dot{\vartheta} = x_2 = f_1(x, u) \\ \dot{x}_2 = \ddot{\vartheta} = -\frac{MgL}{J} \sin(x_1) - \frac{h}{J} x_2 + \frac{1}{J} u = f_2(x, u) \\ y = x_1 = g(x) \end{cases}$$

- Second order
- Nonlinear
- Time-invariant
- Strictly proper
- SISO

State Variables



Quantities associated with **storage** of mass, energy, electrical charge, etc.

- **Electrical** systems:
  - ➔ voltage on capacitors, current in inductors
- **Mechanical** systems:
  - ➔ positions, velocities (linear, angular)
- **Thermal** systems:
  - ➔ temperature, enthalpy
- Etc. ...

Consider a system modelled via the generic differential equation:

$$\frac{d^n y}{dt^n} = \varphi \left( \frac{d^{n-1} y}{dt^{n-1}}, \dots, \frac{dy}{dt}, y, u \right)$$

Letting:

$$\left\{ \begin{array}{l} x_1 := y \\ x_2 := \frac{dy}{dt} \\ \vdots \\ x_n := \frac{d^{n-1} y}{dt^{n-1}} \end{array} \right. \quad \text{and} \quad x := \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \longrightarrow \quad \left\{ \begin{array}{l} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \vdots \\ \dot{x}_n = \varphi(x, u) \\ y = x_1 \end{array} \right.$$

# Example

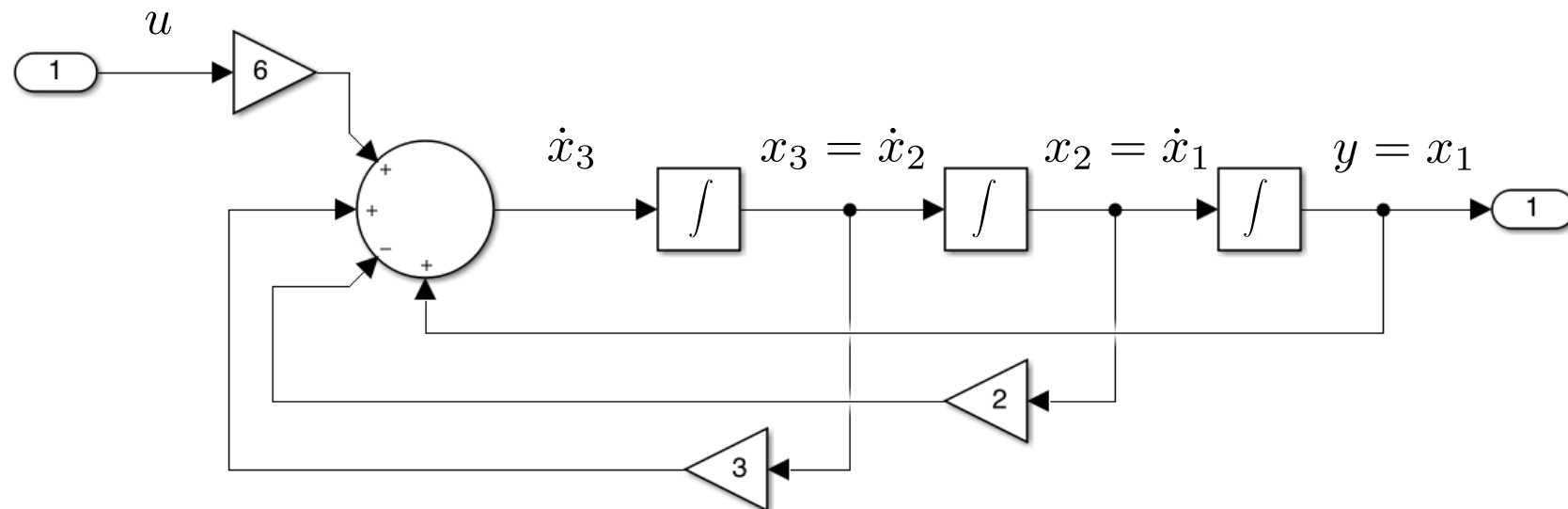


$$\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} - y = 6u$$

[Livescripts in MS Teams](#): see Part 2:  
STATE\_SPACE\_MODEL\_EXAMPLES

Letting: 
$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = 3x_3 - 2x_2 + x_1 + 6u \\ y = x_1 \end{cases}$$

$$\begin{cases} x_1 := y \\ x_2 := \frac{dy}{dt} \\ x_3 := \frac{d^2 y}{dt^2} \end{cases} \quad \text{and} \quad x := \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



# Determination of State and Output Trajectories

$$\begin{cases} \dot{x} = f(x, u, t) \\ y = g(x, u, t) \end{cases} \quad \begin{array}{l} x \in \mathbb{R}^n \\ u \in \mathbb{R}^m \\ y \in \mathbb{R}^p \end{array}$$

$$\left. \begin{array}{l} t_0 \\ x(t_0) \\ u(t), t \geq t_0 \end{array} \right\}$$



state trajectory

$$\overbrace{x(t), t \geq t_0}$$

$$y(t), t \geq t_0$$

output trajectory

a) integration of the state equation   $x(t), t \geq t_0$

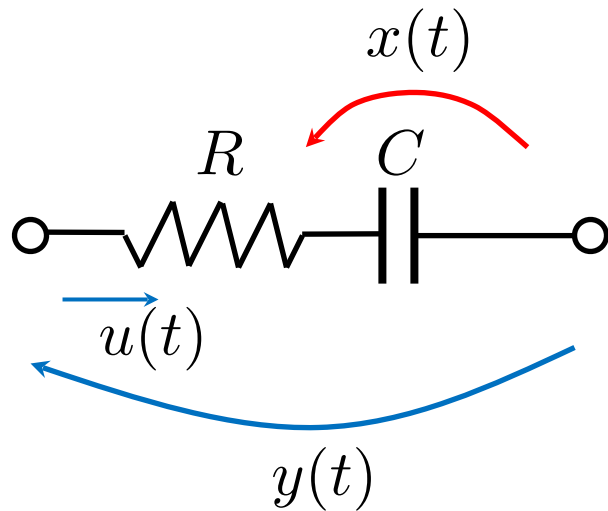
b) substitution of  $x(t), u(t)$  into the output equation   $y(t), t \geq t_0$

**For time-invariant systems we set  $t_0 = 0$  without loss of generality**

# Example 1

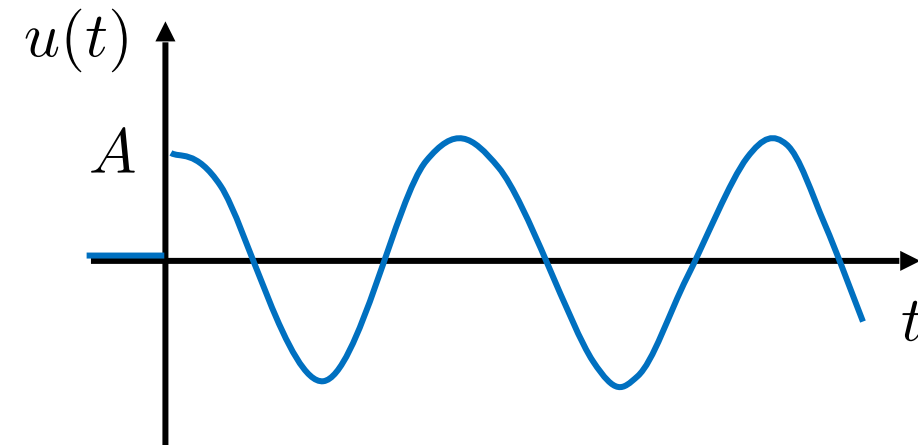


Livescripts in MS Teams: see Part 2:  
STATE\_OUTPUT\_TRAJ\_CONTROL\_TLBX,  
STATE\_OUTPUT\_TRAJ\_LAPLACE



$$\begin{cases} \dot{x}(t) = \frac{1}{C} u \\ y = x + R u \end{cases}$$

$$\begin{aligned} x(0) &= x_0 \\ u(t) &= A \cos(\omega t) \cdot 1(t) \end{aligned}$$



a) Integration of the state equation:

$$\begin{aligned}x(t) &= x_0 + \frac{A}{C} \int_0^t \cos(\omega\tau) d\tau \\&= x_0 + \frac{A}{C} \frac{\sin(\omega\tau)}{\omega} \Big|_0^t \\&= x_0 + \frac{A}{C\omega} \sin(\omega t), t \geq 0\end{aligned}$$

b) Substitution of  $x(t)$ ,  $u(t)$  into the output equation:

$$y(t) = x_0 + \frac{A}{C\omega} \sin(\omega t) + RA \cos(\omega t), t \geq 0$$

The same result can be obtained using the **Laplace Transform**:

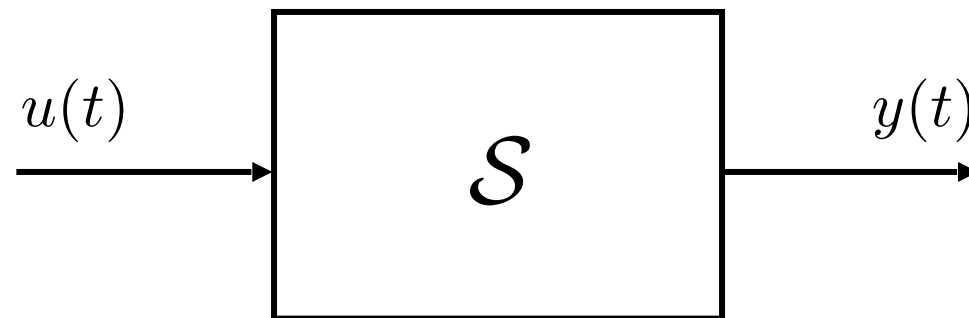
$$\mathcal{L}\{\dot{x}\} = \mathcal{L}\left\{\frac{1}{C}u\right\} \longrightarrow sX(s) - x_0 = \frac{1}{C}U(s)$$

$$\downarrow X(s) = x_0 \frac{1}{s} + \frac{1}{sC}U(s)$$

$$= x_0 \frac{1}{s} + \frac{1}{\cancel{s}C} \frac{\cancel{As}}{s^2 + \omega^2} = x_0 \frac{1}{s} + \frac{A}{C} \frac{1}{s^2 + \omega^2}$$

$$= x_0 \frac{1}{s} + \frac{A}{\omega C} \frac{\omega}{s^2 + \omega^2}$$

$$\mathcal{L}^{-1} \downarrow x(t) = x_0 + \frac{A}{C\omega} \sin(\omega t), t \geq 0$$



$$u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_m(t) \end{bmatrix} \in \mathbb{R}^m$$

Input

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix} \in \mathbb{R}^n$$

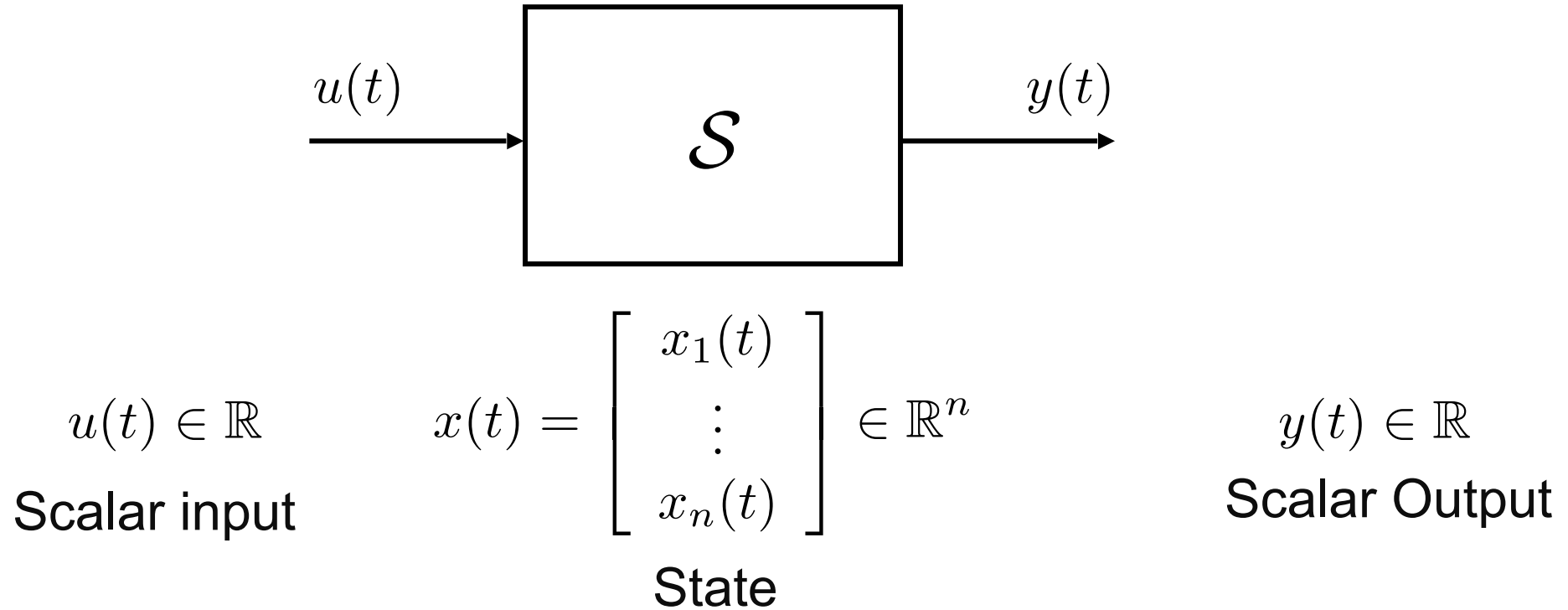
State

$$y(t) = \begin{bmatrix} y_1(t) \\ \vdots \\ y_p(t) \end{bmatrix} \in \mathbb{R}^p$$

Output

$$\begin{cases} \dot{x} = f(x, u) \\ y = g(x, u) \end{cases}$$

System is **linear** if functions  $f(\cdot), g(\cdot)$  **depend linearly** on  $x, u$



System is **linear** if  
functions  $f(\cdot), g(\cdot)$   
**depend linearly**  
on  $x, u$



$$\begin{cases} \dot{x}_1 = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n + b_1u \\ \vdots \\ \dot{x}_n = a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n + b_nu \\ y = c_1x_1 + c_2x_2 + \cdots + c_nx_n + du \end{cases}$$

$$A = \left[ \begin{array}{ccc} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{array} \right] \left. \vphantom{\begin{array}{ccc} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{array}} \right\} n$$

$\underbrace{\hspace{10em}}_n$

$$B = \left[ \begin{array}{c} b_1 \\ \vdots \\ b_n \end{array} \right] \left. \vphantom{\begin{array}{c} b_1 \\ \vdots \\ b_n \end{array}} \right\} n$$

$\underbrace{\hspace{10em}}_1$

$$C = \left[ \begin{array}{ccc} c_1 & \cdots & c_n \end{array} \right] \left. \vphantom{\begin{array}{ccc} c_1 & \cdots & c_n \end{array}} \right\} 1$$

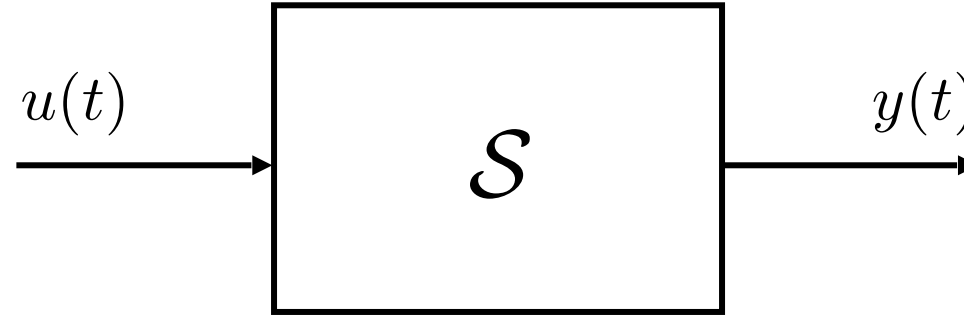
$\underbrace{\hspace{10em}}_n$

$$D = d \in \mathbb{R}$$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad \begin{array}{l} x \in \mathbb{R}^n \\ u \in \mathbb{R} \\ y \in \mathbb{R} \end{array}$$

$(A, B, C, D)$

Compact notation for  
linear systems



$$u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_m(t) \end{bmatrix} \in \mathbb{R}^m$$

Input

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix} \in \mathbb{R}^n$$

State

$$y(t) = \begin{bmatrix} y_1(t) \\ \vdots \\ y_p(t) \end{bmatrix} \in \mathbb{R}^p$$

Output

$$A = \left[ \begin{array}{ccc} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{array} \right] \left. \vphantom{\begin{array}{ccc} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{array}} \right\} n$$

$\underbrace{\hspace{10em}}_n$

$$C = \left[ \begin{array}{ccc} c_{11} & \cdots & c_{1n} \\ \vdots & \ddots & \vdots \\ c_{p1} & \cdots & c_{pn} \end{array} \right] \left. \vphantom{\begin{array}{ccc} c_{11} & \cdots & c_{1n} \\ \vdots & \ddots & \vdots \\ c_{p1} & \cdots & c_{pn} \end{array}} \right\} p$$

$\underbrace{\hspace{10em}}_n$

$$B = \left[ \begin{array}{ccc} b_{11} & \cdots & b_{1m} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nm} \end{array} \right] \left. \vphantom{\begin{array}{ccc} b_{11} & \cdots & b_{1m} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nm} \end{array}} \right\} n$$

$\underbrace{\hspace{10em}}_m$

$$D = \left[ \begin{array}{ccc} d_{11} & \cdots & d_{1m} \\ \vdots & \ddots & \vdots \\ d_{p1} & \cdots & d_{pm} \end{array} \right] \left. \vphantom{\begin{array}{ccc} d_{11} & \cdots & d_{1m} \\ \vdots & \ddots & \vdots \\ d_{p1} & \cdots & d_{pm} \end{array}} \right\} p$$

$\underbrace{\hspace{10em}}_m$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad \begin{array}{l} x \in \mathbb{R}^n \\ u \in \mathbb{R}^m \\ y \in \mathbb{R}^p \end{array} \quad (A, B, C, D)$$

First, for illustration purposes, we consider the case of **scalar state**:

$$\begin{cases} \dot{x} = ax + bu & x \in \mathbb{R} \\ y = cx + du & u \in \mathbb{R} \\ & y \in \mathbb{R} \end{cases} \quad \begin{matrix} x(0) = x_0 \\ u(t), t \geq 0 \end{matrix}$$

By Laplace transformation:

$$\mathcal{L}\{\dot{x}(t)\} = \mathcal{L}\{ax + bu\} \quad \longrightarrow \quad sX(s) - x_0 = aX(s) + bU(s)$$

$$\quad \longrightarrow \quad (s - a)X(s) = x_0 + bU(s) \quad \longrightarrow \quad X(s) = \frac{1}{s - a}x_0 + \frac{b}{s - a}U(s)$$

$$\text{Since: } \mathcal{L}[f(t) * g(t)] = F(s) \cdot G(s); \quad \mathcal{L}[e^{kt}] = \frac{1}{s - k}$$

$$\mathcal{L}^{-1} \quad \longrightarrow \quad x(t) = \mathcal{L}^{-1}\{X(s)\} = e^{at}x_0 + \int_0^t e^{a(t-\tau)}bu(\tau)d\tau, t \geq 0$$

Now, we consider the general case:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad \begin{matrix} x \in \mathbb{R}^n \\ u \in \mathbb{R}^m \\ y \in \mathbb{R}^p \end{matrix} \quad \begin{matrix} x(0) = x_0 \\ u(t), t \geq 0 \end{matrix}$$

By Laplace transformation:

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix} \xrightarrow{\text{green arrow}} X(s) = \mathcal{L}[x(t)] = \begin{bmatrix} X_1(s) \\ \vdots \\ X_n(s) \end{bmatrix}$$

$$\mathcal{L}[Ax(t)] = A\mathcal{L}[x(t)]$$

$$\mathcal{L}[\dot{x}(t)] = \begin{bmatrix} \mathcal{L}[\dot{x}_1(t)] \\ \vdots \\ \mathcal{L}[\dot{x}_n(t)] \end{bmatrix} = \begin{bmatrix} sX_1(s) - x_1(0) \\ \vdots \\ sX_n(s) - x_n(0) \end{bmatrix} = sX(s) - x(0)$$

Hence:

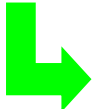
$$\mathcal{L}\{\dot{x}(t)\} = \mathcal{L}\{Ax + Bu\} \quad \longrightarrow \quad sX(s) - x(0) = AX(s) + BU(s)$$

$$\quad \longrightarrow \quad (sI - A)X(s) = x(0) + BU(s)$$

$$\quad \longrightarrow \quad \begin{cases} X(s) &= (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s) \\ Y(s) &= CX(s) + DU(s) \\ &= C(sI - A)^{-1}x(0) + [C(sI - A)^{-1}B + D]U(s) \end{cases}$$

Therefore:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad \begin{matrix} x \in \mathbb{R}^n \\ u \in \mathbb{R}^m \\ y \in \mathbb{R}^p \end{matrix} \quad \begin{matrix} x(0) = x_0 \\ u(t), t \geq 0 \end{matrix}$$


$$x(t) = \mathcal{L}^{-1} \{ (sI - A)^{-1} \} x(0) + \mathcal{L}^{-1} \{ (sI - A)^{-1} B U(s) \}$$

$$= e^{At} x_0 + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau$$

where the **matrix exponential** has been introduced:

$$e^{At} := \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{A^2 t^2}{2} + \dots \quad \text{where} \quad e^{At} = \mathcal{L}^{-1} \{ (sI - A)^{-1} \}$$

Observe that:

$$\begin{aligned}x(t) &= e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \\ &= x_l(t) + x_f(t)\end{aligned}$$

where:

- $x_l(t)$  denotes the **free state trajectory** that only depends on  $x_0$  (**linearly**)
- $x_f(t)$  denotes the **forced state trajectory** that only depends on  $u(t)$  (**linearly**)

$$\begin{aligned}y(t) &= Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}Bu(\tau)d\tau + Du(t) \\ &= y_l(t) + y_f(t)\end{aligned}$$

where:

- $y_l(t)$  denotes the **free output trajectory** that only depends on  $x_0$  (**linearly**)
- $y_f(t)$  denotes the **forced output trajectory** that only depends on  $u(t)$  (**linearly**)

$$\begin{cases} \dot{x} = f(x, u) \\ y = g(x, u) \end{cases} \quad \begin{matrix} x \in \mathbb{R}^n \\ u \in \mathbb{R}^m \\ y \in \mathbb{R}^p \end{matrix} \quad u(t) = \bar{u}, \forall t$$

- **Equilibrium state**

 **Constant** trajectory of  $x(t)$  for a **constant** input  $u(t) = \bar{u}, \forall t$

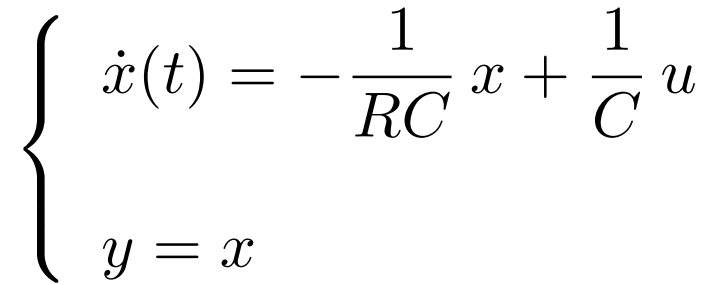
- **All** equilibrium states can be determined selecting inputs  $u(t) = \bar{u} \in \mathbb{R}^m$

- **Equilibrium states**  $\bar{x}$  are the solutions of the **algebraic equation**

$$0 = f(x, \bar{u}) \quad \text{green arrow} \quad \bar{x}$$

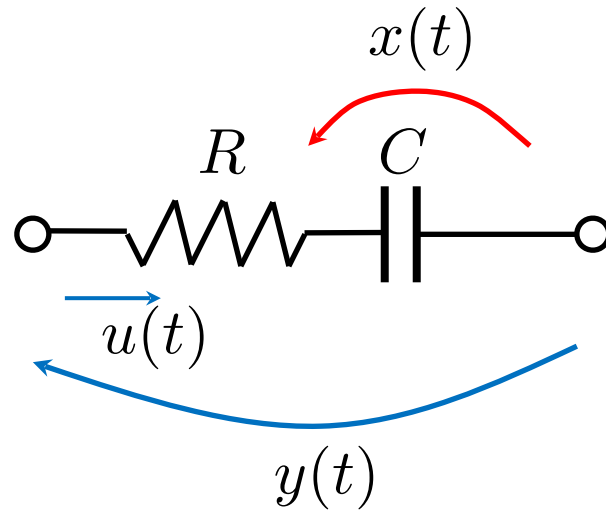
- **Equilibrium outputs** are obtained substituting  $\bar{x}, \bar{u}$  into the output equation

$$\bar{y} = g(\bar{x}, \bar{u})$$


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  $\bar{x} = R\bar{u}$         $\bar{y} = R\bar{u}$

# Equilibrium: Example 2



$$\begin{cases} \dot{x}(t) = \frac{1}{C} u \\ y = x + R u \end{cases}$$

$$u(t) = \bar{u}, \forall t \quad \longrightarrow \quad 0 = \frac{1}{C} \bar{u}$$

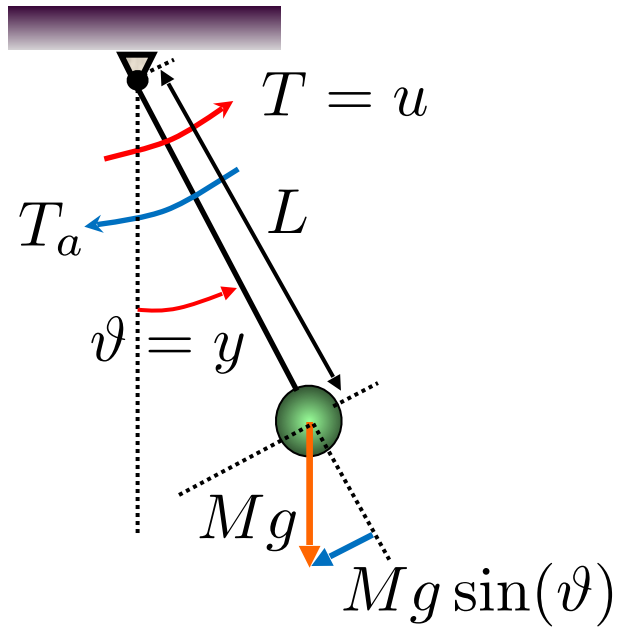
If  $\bar{u} = 0$  

$$\begin{aligned} &\exists \infty \bar{x} \\ &\exists \infty \bar{y} = \bar{x} \end{aligned}$$

If  $\bar{u} \neq 0$  

$$\begin{aligned} &\nexists \bar{x} \\ &\nexists \bar{y} \end{aligned}$$

# Equilibrium: Example 3



$$\begin{cases} x_1 := \vartheta \\ x_2 := \dot{\vartheta} \end{cases}, \quad x := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{cases} \dot{x}_1 = \dot{\vartheta} = x_2 \\ \dot{x}_2 = -\frac{MgL}{J} \sin(x_1) - \frac{h}{J} x_2 + \frac{1}{J} u \\ y = x_1 \end{cases}$$

$$u(t) = \bar{u}, \forall t \quad \longrightarrow \quad \begin{cases} 0 = x_2 \\ 0 = -\frac{MgL}{J} \sin(x_1) - \frac{h}{J} x_2 + \frac{1}{J} \bar{u} \end{cases}$$

$$\begin{cases} \bar{x}_2 = 0 \\ \sin(\bar{x}_1) = \frac{1}{MgL} \bar{u} \end{cases} \quad |\bar{u}| \leq MgL \quad \longrightarrow \quad \begin{cases} \bar{x}_2 = 0 \\ \bar{x}_1 = \arcsin\left(\frac{\bar{u}}{MgL}\right) \end{cases} \quad \longrightarrow \quad \exists \infty \bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}$$

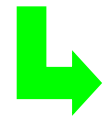
# Equilibria of Linear Time-invariant Systems



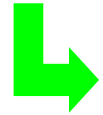
[Livescripts in MS Teams: see  
INTRO\\_MATLAB\\_SIMULINK](#)



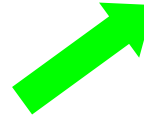
$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad \begin{matrix} x \in \mathbb{R}^n \\ u \in \mathbb{R}^m \\ y \in \mathbb{R}^p \end{matrix} \quad u(t) = \bar{u}, \forall t$$



$$0 = Ax + B\bar{u}$$



$$Ax = -B\bar{u}$$



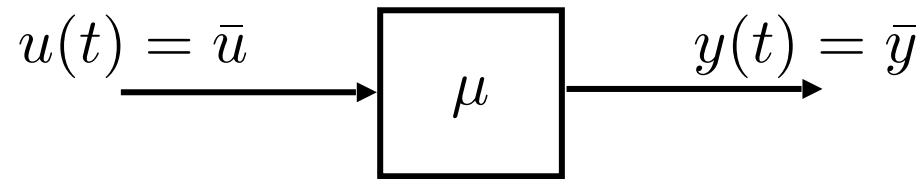
$$\det(A) \neq 0$$



$$\det(A) = 0$$

- $\det(A) \neq 0$    $\bar{x} = -A^{-1}B\bar{u}$  One and only one equilibrium state

  $\bar{y} = C\bar{x} + D\bar{u} = -CA^{-1}B\bar{u} + D\bar{u} = \underbrace{(-CA^{-1}B + D)}_{\text{Static Gain}} \bar{u}$



**Static Gain**

- $\det(A) = 0$    $\exists \infty \bar{x}, \exists \infty \bar{y}$   
  $\nexists \bar{x}, \nexists \bar{y}$

# Equivalent State Equation Representations - Linear case

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad \begin{matrix} x \in \mathbb{R}^n \\ u \in \mathbb{R}^m \\ y \in \mathbb{R}^p \end{matrix}$$

Letting:

$$x = T\hat{x}, \quad T \in \mathbb{R}^{n \times n}, \det(T) \neq 0 \quad \longrightarrow \quad \hat{x} = T^{-1}x$$

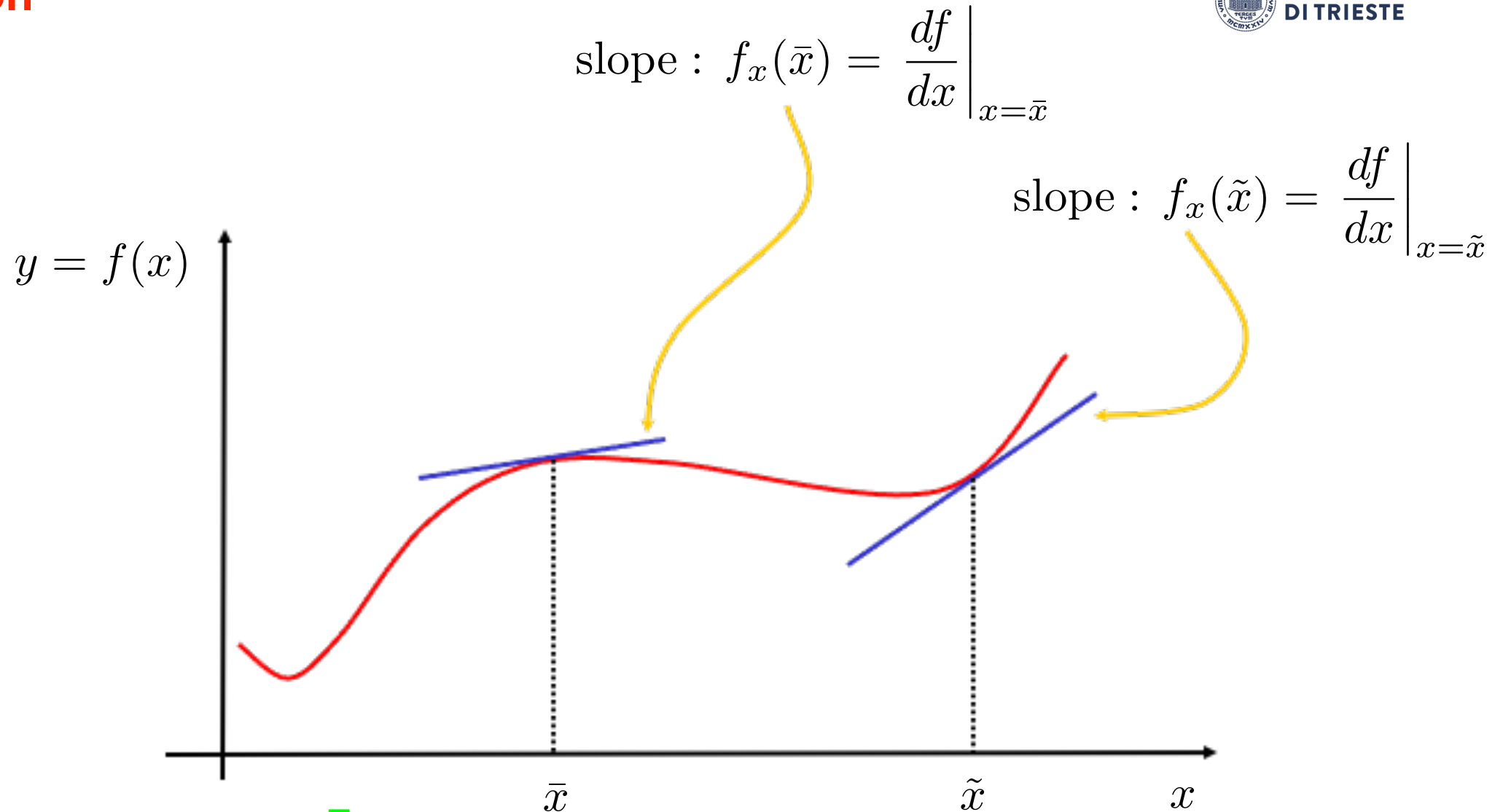
$$\begin{matrix} \downarrow \\ \longrightarrow \end{matrix} \begin{cases} \dot{\hat{x}} = T^{-1}(Ax + Bu) = \underbrace{T^{-1}AT}_{\hat{A}}\hat{x} + \underbrace{T^{-1}B}_{\hat{B}}u \\ y = \underbrace{CT}_{\hat{C}}\hat{x} + Du \end{cases}$$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad \longleftrightarrow \quad \begin{cases} \dot{\hat{x}} = \hat{A}\hat{x} + \hat{B}u \\ y = \hat{C}\hat{x} + Du \end{cases}$$

# Linearisation



Basic Idea:



$$y(x) \simeq y(\bar{x}) + f_x(\bar{x})(x - \bar{x})$$

Deviations from to the equilibrium:

$$\begin{aligned}\delta u(t) &:= u(t) - \bar{u} & u(t) &= \delta u(t) + \bar{u} \\ \delta x(t) &:= x(t) - \bar{x} & \implies x(t) &= \delta x(t) + \bar{x} \\ \delta y(t) &:= y(t) - \bar{y} & y(t) &= \delta y(t) + \bar{y}\end{aligned}$$

♦ State:

$$\begin{aligned}\dot{x} = \delta\dot{x} &= f(\bar{x} + \delta x, \bar{u} + \delta u) \simeq \cancel{f(\bar{x}, \bar{u})} + f_x(\bar{x}, \bar{u})\delta x + f_u(\bar{x}, \bar{u})\delta u \\ &= 0 \quad (\text{equilibrium})\end{aligned}$$



$$\delta\dot{x} \simeq \underbrace{f_x(\bar{x}, \bar{u})\delta x}_{n \times n} + \underbrace{f_u(\bar{x}, \bar{u})\delta u}_{n \times m}$$

$n \times n$

$A$

$n \times m$

$B$

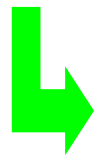
$$A = f_x(\bar{x}, \bar{u}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}_{\bar{x}, \bar{u}}$$

$$B = f_u(\bar{x}, \bar{u}) = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial u_1} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}_{\bar{x}, \bar{u}}$$

◆ Output: = at equilibrium

$$y(t) = \bar{y} + \delta y = g(\bar{x} + \delta x, \bar{u} + \delta u) \simeq g(\bar{x}, \bar{u}) + g_x(\bar{x}, \bar{u})\delta x + g_u(\bar{x}, \bar{u})\delta u$$

$$C = g_x(\bar{x}, \bar{u}) = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial g_p}{\partial x_1} & \dots & \frac{\partial g_p}{\partial x_n} \end{bmatrix}_{\bar{x}, \bar{u}}$$



$$\delta y \simeq \underbrace{g_x(\bar{x}, \bar{u})\delta x}_{\substack{p \times n \\ C}} + \underbrace{g_u(\bar{x}, \bar{u})\delta u}_{\substack{p \times m \\ D}}$$

$$D = g_u(\bar{x}, \bar{u}) = \begin{bmatrix} \frac{\partial g_1}{\partial u_1} & \dots & \frac{\partial g_1}{\partial u_m} \\ \vdots & & \vdots \\ \frac{\partial g_p}{\partial u_1} & \dots & \frac{\partial g_p}{\partial u_m} \end{bmatrix}_{\bar{x}, \bar{u}}$$

Summing up:

$$\begin{cases} \dot{x} = f(x, u) \\ y = g(x, u) \end{cases} \implies 0 = f(x, \bar{u}) \implies \bar{x} \text{ (equilibrium state)}$$

$$\delta \dot{x} \simeq \underbrace{f_x(\bar{x}, \bar{u})}_{A} \delta x + \underbrace{f_u(\bar{x}, \bar{u})}_{B} \delta u$$

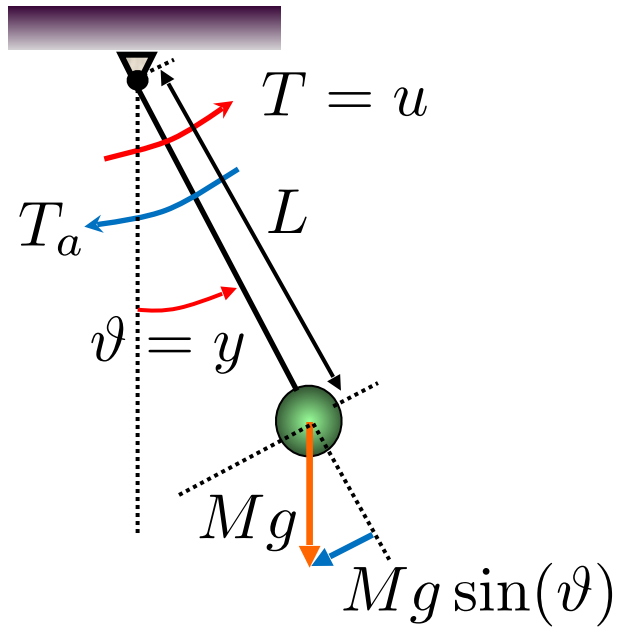


$$\delta y \simeq \underbrace{g_x(\bar{x}, \bar{u})}_{C} \delta x + \underbrace{g_u(\bar{x}, \bar{u})}_{D} \delta u$$



$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} ; \begin{array}{l} x \in \mathbb{R}^n \\ u \in \mathbb{R}^m \\ y \in \mathbb{R}^p \end{array}$$

# Example



Letting (see slide 27):

$$\begin{cases} x_1 := \vartheta \\ x_2 := \dot{\vartheta} \end{cases}, \quad x := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{cases} \dot{x}_1 = \dot{\vartheta} = x_2 \\ \dot{x}_2 = -\frac{MgL}{J} \sin(x_1) - \frac{h}{J} x_2 + \frac{1}{J} u \\ y = x_1 \end{cases}$$

$$u(t) = \bar{u} = 0, \forall t \quad \longrightarrow \quad \begin{cases} 0 = x_2 \\ 0 = -\frac{MgL}{J} \sin(x_1) - \frac{h}{J} x_2 \end{cases}$$

$$\begin{cases} \bar{x}_2 = 0 \\ \sin(\bar{x}_1) = \frac{1}{MgL} \bar{u} \end{cases} \quad \longrightarrow \quad \begin{cases} \bar{x}_2 = 0 \\ \bar{x}_1 = k\pi, \quad \forall k \in \mathbb{Z} \end{cases}$$

We pick the two "physical" solutions:

$$\bar{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}; \quad \tilde{x} = \begin{bmatrix} \pi \\ 0 \end{bmatrix}$$

We get:

$$f_x(\bar{x}, \bar{u}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{\bar{x}, \bar{u}} = \begin{bmatrix} 0 & 1 \\ -\frac{MgL}{J} \cos(x_1) & -\frac{h}{J} \end{bmatrix}_{\bar{x}, \bar{u}} = \begin{bmatrix} 0 & 1 \\ -\frac{MgL}{J} & -\frac{h}{J} \end{bmatrix} = \bar{A}$$

$$f_x(\tilde{x}, \bar{u}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{\tilde{x}, \bar{u}} = \begin{bmatrix} 0 & 1 \\ -\frac{MgL}{J} \cos(x_1) & -\frac{h}{J} \end{bmatrix}_{\tilde{x}, \bar{u}} = \begin{bmatrix} 0 & 1 \\ +\frac{MgL}{J} & -\frac{h}{J} \end{bmatrix} = \tilde{A}$$

$$f_u(\bar{x}, \bar{u}) = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\bar{x}, \bar{u}} = \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix} = \bar{B} ; \quad f_u(\tilde{x}, \bar{u}) = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\tilde{x}, \bar{u}} = \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix} = \tilde{B}$$

$$g_x(\bar{x}, \bar{u}) = \begin{bmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} \end{bmatrix}_{\bar{x}, \bar{u}} = \begin{bmatrix} 1 & 0 \end{bmatrix} = \bar{C} ; \quad g_x(\tilde{x}, \bar{u}) = \begin{bmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} \end{bmatrix}_{\tilde{x}, \bar{u}} = \begin{bmatrix} 1 & 0 \end{bmatrix} = \tilde{C}$$

$$g_u(\bar{x}, \bar{u}) = \left. \frac{\partial g}{\partial u} \right|_{\bar{x}, \bar{u}} = 0 = \bar{D} ; \quad g_u(\tilde{x}, \bar{u}) = \left. \frac{\partial g}{\partial u} \right|_{\tilde{x}, \bar{u}} = 0 = \tilde{D}$$